

WORKING DRAFT

August 6, 2026

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Todo list



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**Aligning Large Language Models to
publicly collected values**

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Submitted in partial fulfilment of the requirements for
Master of Artificial Intelligence.

Abstract

A short description of the project goes here.

Acknowledgments

Any acknowledgments should go in here, between the title page and the table of contents. The acknowledgments do not form a proper chapter, and so don't get a number or appear in the table of contents.

Contents

1	Introduction	1
1.1	Report overview	1
2	Background	3
3	Fine-tuning	5
3.1	Fine-tuning on diverse clusters with appreciable training	5
3.1.1	The clusters of survey respondents	5
3.1.2	Efficacy of fine-tuning	5
3.2	Creating the training dataset	5
3.2.1	Clustering of survey data	5
3.2.2	Structure of the training data	6
3.3	Fine-tuning methods	6
3.3.1	SFT fine-tuning: Modal Sampled response	6
3.3.2	LLM models	6
3.3.3	Supervised fine tuning loss function	7
3.3.4	Distributional loss functions	7
4	Simulation	9
5	Survey	11
6	Conclusions	13
A	Project details	17
A.1	Budget and Resources	17

Figures

Chapter 1

Introduction

This chapter gives an introduction to the project report.

5-6 pages with a full summary

1.1 Report overview

Following on from this introduction chapter there are 5 more chapters to the report. The next chapter 2 gives the key background

This report is structured into 3 main chapters, each focusing on a specific part of the project. For each chapter of the project it explains the methods used and the results obtained. Each chapter is self contained and can be read somewhat independently of the others. The 3 main chapters are:

- 3: Fine-tuning of the LLM on WVS data
- 4: Behavioural analysis of the fine-tuned and base LLMs
- 5: Public consultation on the alignment of the LLMs

The final chapter 6 concludes the report with a summary of the findings, conclusions drawn from the project, key limitations and recommendations for future work.

Chapter 2

Background

A related works section.

Could be removed from word count.

Includes a biref (few pages) summary of the important bits of the related works and how they relate to this project.

This work can understand from multiple perspectives. Here are some
survey modelling [1] [3]
annotation disagreement [2]
student teacher and distillation

Chapter 3

Fine-tuning

This chapter will talk about the fine-tuning process used.

It will mention clustering of WVS data prompts and format of the data finetuning methods finetuning evaluations (survey modelling and response distributions)

The goal of this chapter is to show that an open weight model can be fine-tuned with value survey data to produce a model that responds more like the respondents it is aligned to when answering survey questions.

In section 3.1 we will find that ...

In section 3.2 and section 3.3 I describe the methods used to create the dataset and fine-tune the model.

3.1 Fine-tuning on diverse clusters with appreciable training

3.1.1 The clusters of survey respondents

To test for inter group differences we need to cluster the survey respondents.

The clusters are distinct

We can assign to the clusters with a high level of confidence

3.1.2 Efficacy of fine-tuning

Discuss the various methods of evaluating the efficacy of fine-tuning, including: - Survey modelling (how well the model predicts survey responses) - Response distributions (how well the model's response distributions match the survey data)

Also explore other concepts of: - Is fine-tuning robust to reasoning on and off? - Does system prompt matter?

3.2 Creating the training dataset

We choose the WVS dataset (free, open and sufficient).

3.2.1 Clustering of survey data

We did clustering using LCA method

This is what the clustering looked like...

3.2.2 Structure of the training data

System prompts and user prompts, how they were selected.

3.3 Fine-tuning methods

To better understand the effects on downstream behaviour 2 types of fine-tuning were done. First is modal fine-tuning where the goal is to have a model that responds in a way that is like the most common response in the survey data for that sub population. The second is distributional fine-tuning where the goal is to have a model that responds in a way that is like the distribution of responses in the survey data for that sub population.

Given the use of LLM reasoning in downstream applications allows for more uncertainty (citation needed, i.e the intuition that the reasoning gives it more chance of randomness and therefore can better model uncertainty).

There are ways of doing finetuning of a LLM. We will only explore simple fine-tuning methods (SFT) and vary the method on two axes: what the target is (modal response, sampled response or the distribution of responses) and what the response representation is (full option string or just single token representing the option).

The modal response fine tuning is conducted as a baseline as it is a straight forward application of SFT. The rest of the methods are various ways of trying to have the model learn the distribution of responses in the survey data. However in an implementation sense there are three methods (SFT, full option string distributional fine-tuning and single token distributional fine-tuning).

3.3.1 SFT fine-tuning: Modal Sampled response

To do modal fine-tuning the model is trained used SFT with the expected text response being the full option string of a response. For modal fine-tuning the expected text response is the most common response in the survey data for that sub population. For sampled response fine-tuning the expected text response is a randomly sampled response from the survey data for that sub population.

[ouyangTrainingLanguageModels2022] shows SFT used before RLHF

[zhouLIMALessMore2023] shows that SFT is good enough.

3.3.2 LLM models

Definitions: V is the vocabulary size, The model parameters are θ . The model is a function f_θ that maps a prompt and a completion prefix to logits over the vocabulary.

Let a training example consist of a prompt (system + user) tokenised as $x = (x_1, \dots, x_p)$ and a completion (assistant answer) tokenised as $y = (y_1, \dots, y_T)$, with T the completion length.

The model is autoregressive, so the probability of the completion factorises over tokens:

$$p_\theta(y \mid x) = \prod_{t=1}^T p_\theta(y_t \mid x, y_{<t}), \quad y_{<t} = (y_1, \dots, y_{t-1}). \quad (3.1)$$

The probability of the next token is given by a softmax over the logits:

$$p_\theta(y_t \mid x, y_{<t}) = \frac{\exp f_\theta(x, y_{<t})_{y_t}}{\sum_{v=1}^V \exp f_\theta(x, y_{<t})_v}. \quad (3.2)$$

3.3.3 Supervised fine tuning loss function

SFT fits the model by *maximum likelihood* on the demonstration data: maximising $\log p_\theta(y \mid x)$ is equivalent to minimising the negative log-likelihood (NLL),

$$\mathcal{L}_{\text{SFT}}(\theta; x, y) = -\log p_\theta(y \mid x) = -\sum_{t=1}^T \log p_\theta(y_t \mid x, y_{<t}), \quad (3.3)$$

and the objective over the dataset $\mathcal{D}$ is $\theta^* = \arg \min_{\theta} \sum_{(x,y) \in \mathcal{D}} \mathcal{L}_{\text{SFT}}(\theta; x, y)$.

Each per-token term in (3.3) is a categorical cross-entropy: with $z_t = f_\theta(x, y_{<t}) \in \mathbb{R}^V$ the logits over the vocabulary of size V ,

$$-\log p_\theta(y_t \mid x, y_{<t}) = -z_{t,y_t} + \log \sum_{v=1}^V \exp(z_{t,v}). \quad (3.4)$$

3.3.4 Distributional loss functions

Rather than modelling a single completion y instead we can model distributions over the K possible completions of a prompt¹. Therefore the objective is to minimise the difference between the models predicted distribution p and the expected distribution q .

$$\mathcal{L}_{\text{distributional}}(\theta; x, y) = -\sum_{i=1}^K q_i * \log \tilde{p}_\theta(y_i \mid x) \quad (3.5)$$

where $\tilde{p}_i$ is the softmax over the K completions of the options log-probs.

$$\tilde{p}_\theta(y_i \mid x) = \frac{\exp(\log p_\theta(y_i \mid x))}{\sum_{j=1}^K \exp(\log p_\theta(y_j \mid x))} \quad (3.6)$$

There are two ways in which we use this loss function. The first is that y is a full string of the response and the second is that y is a single token representing the response.

We can see that the distributional loss function is a generalisation of the SFT loss function. If we have a single response then $K = 1$ and the distributional loss function reduces to the SFT loss function (3.3) as $q_i = 1$ for the expected y and 0 elsewhere.

Full option string distributional fine-tuning

When $T > 1$

Single token distributional fine-tuning

This is the simpler of the two and used in other research [2, 1]...

¹This is possible as the task is a classification task with a finite number of possible completions.

Chapter 4

Simulation

A section discussion the behavioural simulations of the model

The selection of scenarios The harness used How summaries of the results were generated
A method for mapping to value dimensions The results of the simulations (compared to clusters are there behavioural clusters)

Chapter 5

Survey

A discussion of the survey

How the questions were selected for assign to cluster

How the scenarios were chosen

Sampling methods

Finding of how well aligned the model is

Chapter 6

Conclusions

The conclusions are presented in this Chapter.

Bibliography

- [1] Yong Cao et al. *Specializing Large Language Models to Simulate Survey Response Distributions for Global Populations*. <https://arxiv.org/abs/2502.07068v2>. Feb. 2025. (Visited on 08/04/2026).
- [2] Tommaso Fornaciari et al. “Beyond Black & White: Leveraging Annotator Disagreement via Soft-Label Multi-Task Learning”. In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Ed. by Kristina Toutanova et al. Online: Association for Computational Linguistics, June 2021, pp. 2591–2597. DOI: 10.18653/v1/2021.naacl-main.204. (Visited on 08/04/2026).
- [3] Ji Huang, Mengfei Li, and Shuai Shao. *Distribution Shift Alignment Helps LLMs Simulate Survey Response Distributions*. <https://arxiv.org/abs/2510.21977v2>. Oct. 2025. (Visited on 08/04/2026).

Appendix A

Project details

A.1 Budget and Resources

Funding (received)	
Blue dot impact rapid grant	2542
VUW programme	500
VUW AI centre	1000
Total funding	4042
Expenses	
GPU compute (160 hours at \$2.00/hour)	320
Participant compensation (500 participants at \$5.25 each)	2625
Participant recruitment admin fee	500
Contingency (20% of direct costs)	689
Total expenses	4134
Balance (funding – expenses)	-92

Table A.1: Detailed project budget in New Zealand Dollars (NZD).